Video game physics: All smoke and mirrors?

Or: Can game engine be used as scientific tools?

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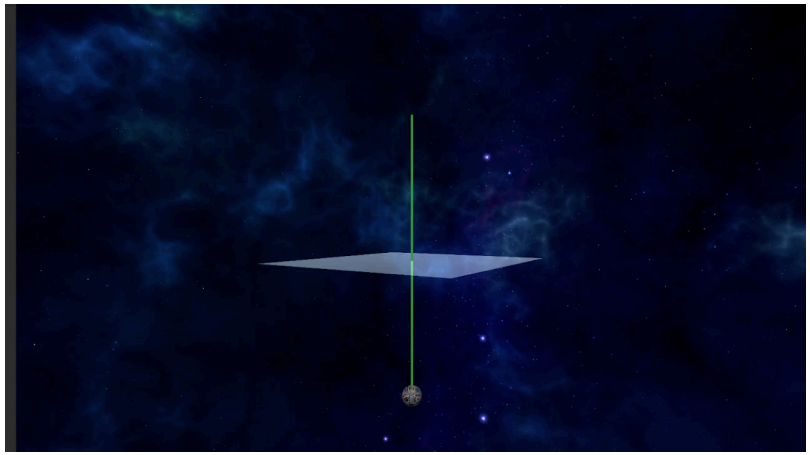
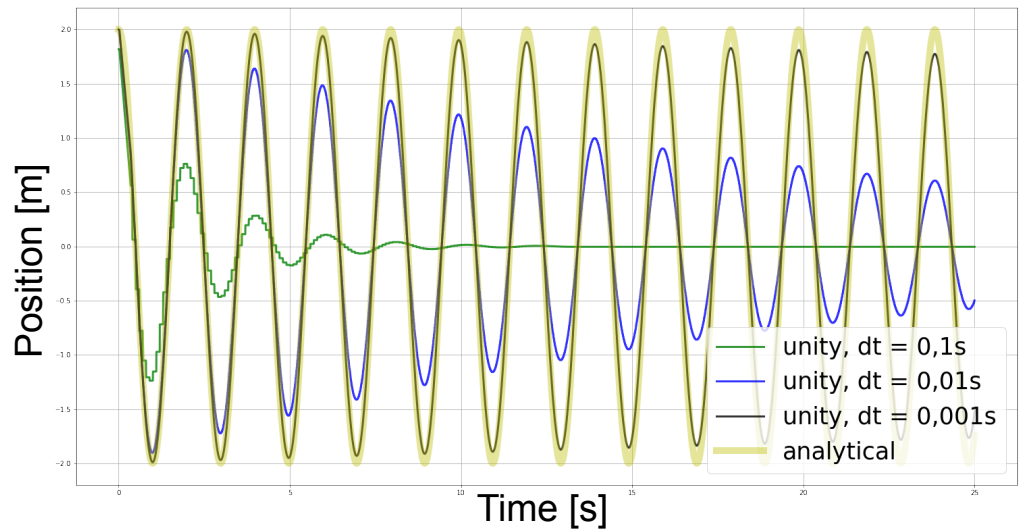
Overview

- The question: Can game engine be used as scientific tools? / how precise is game physics?
- What is Unity engine?
- Why do we investigate the physics capabilities of a game engine?
- How do we quantify how “precise” this physics engine is?

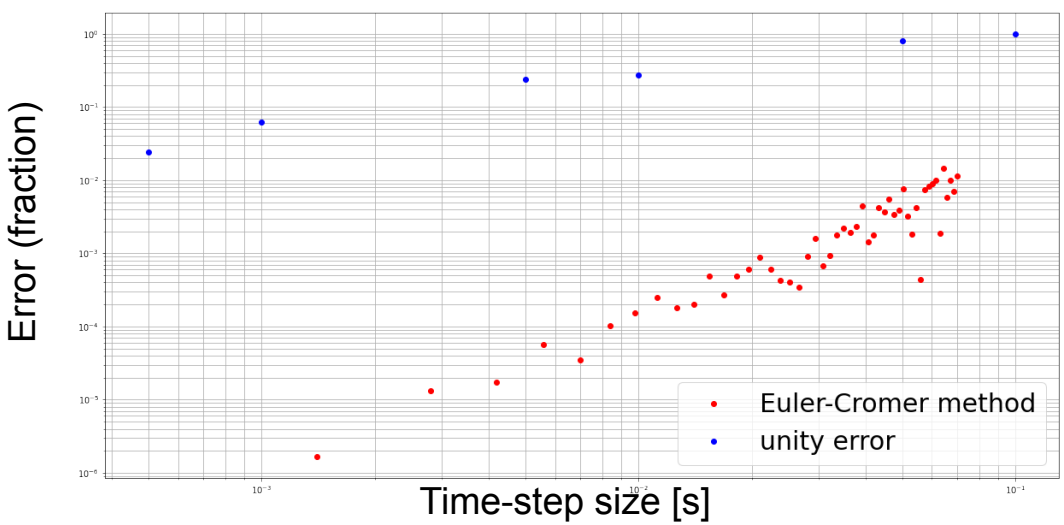
Simple harmonic oscillator



Plot of the position vs. time



Accuracy of the 6th peak (t ≈ 10[s])



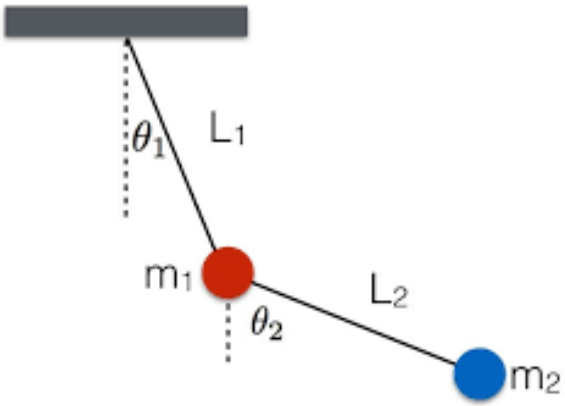
- Initial conditions:

$$y_0 = 2[m],\ y_{rest} = 0,\ k = 10\left[\frac{N}{m}\right],\ L = 10[m]$$

- Unity's physics engine will dissipate energy during time
- Accuracy of unity behaves logarithmically
- Unable to run below dt = 0.001s due to lack of computational power

Double pendulum: Modelling

- The Double Pendulum is famously known as a *chaotic* system i.e its path changes greatly with initial conditions



$$V = -(m_1 + m_2)L_1 * g * cos(\theta_1) - m_2 * L_2 * g * cos(\theta_2)$$

$$T = \frac{1}{2}m_1 * L_1^2 * \dot{\theta_1}^2 + \frac{1}{2}m_2[L_1^2 * \dot{\theta_1}^2 + L_2^2 * \dot{\theta_2}^2 + 2 * L_1 * L_2 * \dot{\theta_1} * \dot{\theta_2} * cos(\theta_1 - \theta_2)]$$

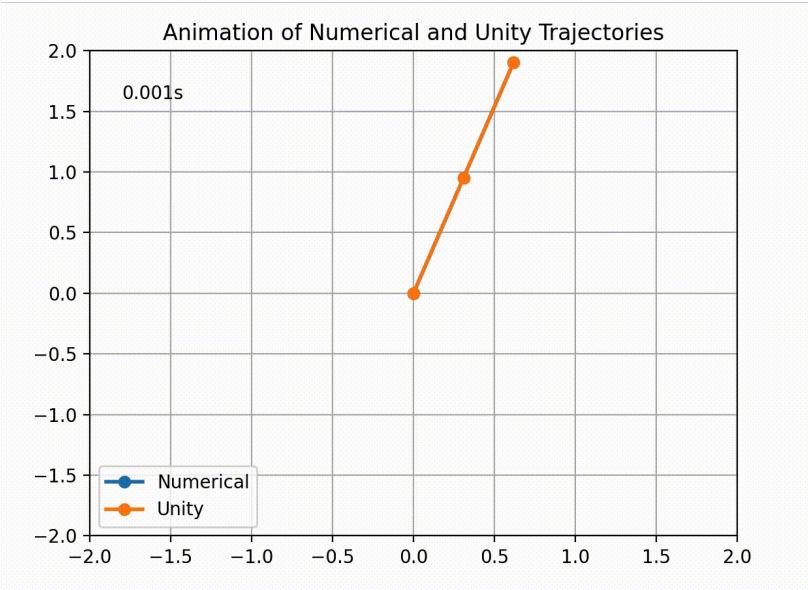
$$L = T - V$$

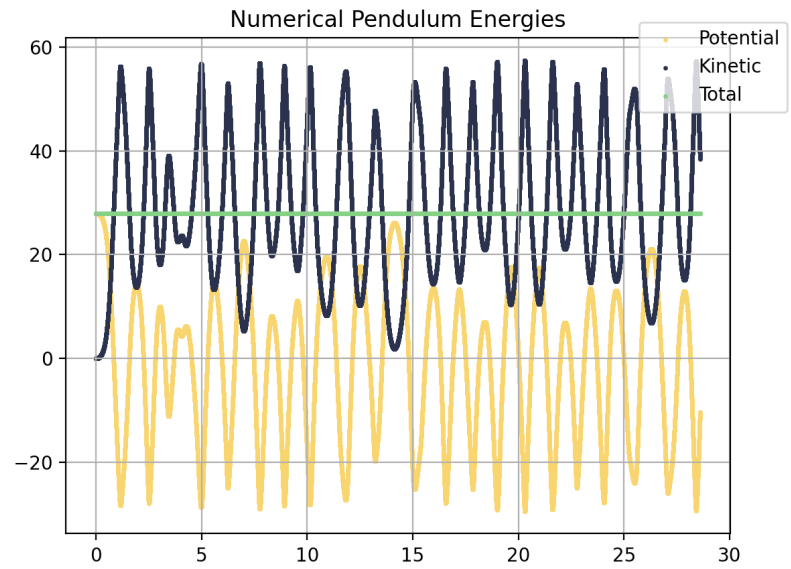
- The positions can then be found by finding the solutions to the pair of ODE's

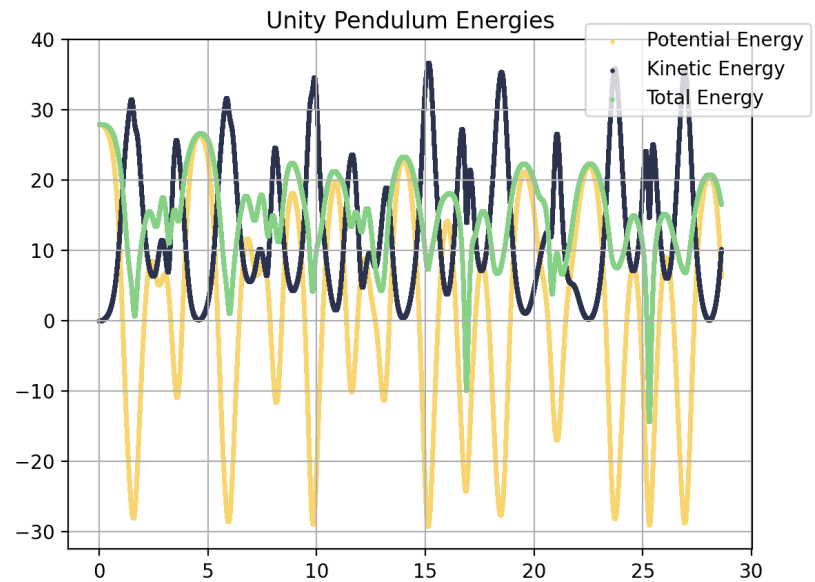
$$\frac{d}{dt}(\frac{\partial L}{\partial q_i}) - \frac{\partial L}{\partial q_i} = 0$$

where $q_1 = \theta_1$ and $q_2 = \theta_2$

Double Pendulum: Results







Double pendulum in Unity

Conclusions: *Can Unity help us in our scientific work?*

- Unity for scientific calculations: **NO**
  - Does not accurately portray motion in real-life
  - Energies are not conserved within the simulation

- Unity for visualisation of scientific results: **YES**
  - Scripts written in C# - efficient
  - Can write scripts to compensate for inaccuracies eg. inherent damping
  - Results can be easily shared